

BINAY—Lighting Up The Future

Binay LED Aviation Obstruction Lights



**BINAY LED-based MEDIUM INTENSITY
Aviation Obstruction Light Beacon**

As per the International Civil Aviation Organisation (ICAO) requirements for use on structures above 45m

Available in Low Intensity and Medium Intensity versions (as per the International Civil Aviation Organization guidelines), BINAY's patented LED Aviation Lights come with a 5-year warranty

LED Obstruction Lighting for:

- Industrial chimneys and smokestacks
- Transmission, microwave, and cellular towers
- Radio, TV and similar structural towers
- High-rise buildings and structures
- Airports and airfields

The BINAY LED Aviation Obstruction Light offers the following advantages:

- Fit-and-forget maintenance-free operation
- A long life of 100,000 hours (20 years at 12 hours daily burning)
- Pays for itself within a short period of operation in the form of reduced installation, maintenance and servicing costs
- Quick installation; Reliable operation 365 days per year
- Shock-proof and vibration-resistant
- Over-Designed Intensity to allow for natural LED intensity degradation over its operating lifetime



**THE BINAY LED OBSTRUCTION LIGHT IS UNDER ACCEPTED
PATENT, AND AS SUCH IS A PROPRIETARY PRODUCT**

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Nano ear that can hear bacteria

Physicists at the Nanosystems Initiative Munich (NIM) have created a nanoear that can detect sound levels as low as -60 dB.

The nanoear is a single gold nanoparticle kept in a state of levitation by a laser beam. Upon weak acoustic excitation, the particle oscillates parallel to the direction of sound propagation.

To test the nano-ear, the researchers placed the gold particle, which is just 60 nanometres across, inside the laser beam. They then took some other microscopic gold particles and heated them using a laser. The sound vibrations given off by this heating then traveled past the nano-ear. The movement of the particle revealed not just the exact decibel level created by the heated particles, but also from what direction they had come.

Admittedly, the nano-ear only worked when placed in a carefully controlled, perfectly arranged environment. For it to be of any use as an acoustic microscope, the researchers will need to come up with a way for the nano-ear to distinguish between sound vibrations and the occasional random movement by the trapped molecule.

Sunflower-inspired solar plants

Researchers at MIT, in collaboration with RWTH Aachen University in Germany, have come up with a design that reduces the land requirement to build a concentrated solar power (CSP) plant, while increasing the amount of sunlight its mirrors collect.

A computational model was developed to evaluate the efficiency of heliostat layouts. The model was then tested on an existing commercial-scale CSP plant—PS10. The researchers tinkered with the pattern of heliostats using numerical optimisation to first bring the fanned-out layout closer together. The narrower layout reduced the amount of land the mirrors took up by 10 per cent without affecting the mirrors' efficiency in reflecting light.

The researchers devised a spiral field with its heliostats rearranged to resemble a sunflower, with each mirror angled about 137 degrees relative to its neighbour. The numerically optimised layout takes up 20 per cent less space than the PS10 layout. What's more, the spiral pattern reduced shading and blocking and increased total efficiency compared to PS10's radially staggered configuration.

Marine solar cells

Can you imagine a solar cell taking energy from water? Phil Pauley, founder of design and innovation agency PAULEY, has developed a marine solar cell—a hybrid solar and wave energy generator with the ability to establish photovoltaic power generation off-shore.

The solar-wave unit captures both wave energy through natural buoyancy displacement and solar energy through photovoltaic cells as natural sunlight also reflects from the surface water to increase solar capture by 20 per cent. This contrasts with conventional solar farms or wave power designs which only harvest one form of power.